# 18 CHEMICAL TESTS

In this chapter we will look at how to identify substances using chemical tests.



▲ Figure 18.1 Although the reactions we learn here are useful for identifying substances in the lab and we are also introduced to some interesting chemistry, analysis of substances is now routinely carried out using sophisticated machines.

## LEARNING OBJECTIVES

- Describe tests for these gases:
  - hydrogen
  - oxygen
  - carbon dioxide
  - ammonia
  - chlorine
- Describe how to carry out a flame test
- Know the colours formed in flame tests for these cations:
  - $Li^+$ (red)
  - $Na^+$ (yellow)
  - $K^+$ (lilac)
  - $Ca^{2+}$ (orange-red)
  - $Cu^{2+}$ (blue-green)
- Describe tests for these cations:
  - $NH_4^+$ using sodium hydroxide solution and identifying the gas evolved
  - $Cu^{2+}$, $Fe^{2+}$ and $Fe^{3+}$ using sodium hydroxide solution
- Describe tests for these anions:
  - $Cl^-$, $Br^-$ and $I^-$ using acidified silver nitrate solution
  - $SO_4^{2-}$ using acidified barium chloride solution
  - $CO_3^{2-}$ using hydrochloric acid and identifying the gas evolved
- Describe a test for the presence of water using anhydrous copper(II) sulfate
- Describe a physical test to show whether or not a sample of water is pure

# TESTING FOR GASES

## HYDROGEN, $H_2$

### THE TEST FOR HYDROGEN GAS

A lighted splint is held to the mouth of the tube. The hydrogen explodes with a squeaky pop.

The hydrogen combines explosively with oxygen in the air to make water.

$$2H_2(g) + O_2(g) \rightarrow 2H_2O(l)$$

## OXYGEN, $O_2$

### THE TEST FOR OXYGEN GAS

A glowing splint is put into the tube containing the gas. Oxygen relights a glowing splint.

## CARBON DIOXIDE, $CO_2$

### THE TEST FOR CARBON DIOXIDE GAS

The carbon dioxide is bubbled through limewater. Carbon dioxide turns limewater milky/chalky/cloudy.

Limewater is calcium hydroxide solution. Carbon dioxide reacts with it to form a white precipitate of calcium carbonate.

$$Ca(OH)_2(aq) + CO_2(g) \rightarrow CaCO_3(s) + H_2O(l)$$

**KEY POINT**

If you keep bubbling the carbon dioxide through the limewater, the white precipitate will eventually disappear again. This is because the $CO_2$ reacts with the calcium carbonate to form calcium hydrogencarbonate, which is soluble in water:

$$CaCO_3(s) + H_2O(l) + CO_2(g) \rightarrow Ca(HCO_3)_2(aq)$$

## CHLORINE, $Cl_2$

### THE TEST FOR CHLORINE GAS

A piece of damp litmus paper or universal indicator paper is put into the test-tube or held over its mouth. Chlorine is a green gas that bleaches (turns white) the damp litmus paper or universal indicator paper.

If blue litmus paper or universal indicator paper are used for this test, they go red first (the chlorine dissolves in the water to form an acidic solution) and then white.

**!**

Safety Note: Wear eye protection. Avoid inhaling or 'sniffing' chlorine or ammonia, especially if you have a breathing problem such as asthma.



▶ Figure 18.2 Chlorine bleaches damp litmus paper.

## AMMONIA, $NH_3$

### THE TEST FOR AMMONIA GAS

Hold a piece of damp universal indicator paper or red litmus paper at the mouth of the test-tube. Ammonia turns the universal indicator paper/litmus paper blue.

Ammonia is the only alkaline gas that you will meet at International GCSE.

# TESTING FOR WATER

## USING ANHYDROUS COPPER(II) SULFATE

Water turns white anhydrous copper(II) sulfate blue.

Anhydrous copper(II) sulfate lacks water of crystallisation and is white. Dropping water onto it replaces the water of crystallisation and turns it blue.

anhydrous copper sulfate + water → hydrated copper sulfate

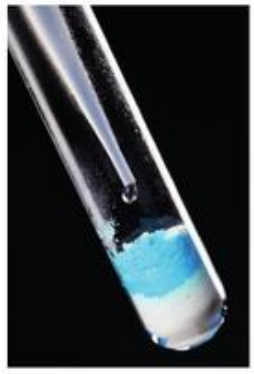
$CuSO_4(s)$ + $5H_2O(l)$ → $CuSO_4 \cdot 5H_2O(s)$

white blue

If you aren't sure about water of crystallisation, see Chapter 17.

This test works for anything that contains water, so would work with sodium chloride solution or sulfuric acid. It does *not* show that the water is pure.

! Safety Note: Avid skin contact with the copper salts.



▲ Figure 18.3 Testing for water with anhydrous copper(II) sulfate.

## PHYSICAL TEST TO SHOW THAT WATER IS PURE

You can find out whether a sample of water is pure by measuring the freezing (melting) point or boiling point. Pure water freezes at exactly 0 °C and boils at exactly 100 °C at 1 atmosphere pressure. If the water is impure, it will usually freeze at a lower temperature and boil at a higher temperature.

# TESTING FOR IONS

All salts contain at least one cation (positive ion) and anion (negative ion). In this section we will learn how to test for these anions and cations. We do the tests for these anions and cations separately.

## FLAME TESTS

A flame test is used to show the presence of certain metal ions (cations) in a compound. A platinum or nichrome wire is dipped into concentrated hydrochloric acid and then into the salt you want to test, so that some salt sticks on the end. The wire and the salt are then held just within a non-luminous (roaring) Bunsen burner flame and the colour observed.

**KEY POINT**

If a new piece of wire is not being used, it is usually cleaned by dipping it into concentrated hydrochloric acid and then holding it in a non-luminous Bunsen flame. This is repeated until the wire doesn't give any colour to the flame.

**HINT**

The colour for sodium is described as yellow in the specification, but you will also see it described as orange. Both are usually acceptable.

**KEY POINT**

Lithium isn't the only ion to give a red flame colour. Strontium gives a very similar colour. The only way to be sure is to compare the flame colour next to the flame colour of a known compound of lithium or strontium.

**!**

Safety Note: Wear eye protection and take great care with the concentrated acid. Do not get it on your skin and watch out for 'spitting' when heating it on the wire.



▲ Figure 18.4 Red shows lithium ions.



▲ Figure 18.5 Yellow shows the presence of sodium ions.



▲ Figure 18.6 Lilac shows potassium ions.



▲ Figure 18.7 Orange-red shows calcium ions.



▲ Figure 18.8 Blue-green shows the presence of copper(II) ions.

## TESTING FOR CATIONS (POSITIVE IONS) USING SODIUM HYDROXIDE SOLUTION

Of the common hydroxides, only sodium, potassium and ammonium hydroxides dissolve in water to any extent. Most metal hydroxides are insoluble. That means that if you add sodium hydroxide solution to a solution containing the metal ions, you will get a precipitate of the metal hydroxide.

To carry out this test, you dissolve the salt in distilled (pure) water and put about 1 $cm^3$ in a test-tube. Add about the same volume of dilute sodium hydroxide solution. What you are looking for in most of these tests is the formation of a precipitate; the solution will go cloudy as a solid forms.

### THE TEST FOR COPPER(II) IONS

**HINT**

Ionic equations, which just show the things that actually react, can be a useful way of showing what is happening in a precipitation reaction.

Formation of a blue precipitate shows the presence of copper(II) ions. The precipitate is copper(II) hydroxide.

$$Cu^{2+}(aq) + 2OH^-(aq) \rightarrow Cu(OH)_2(s)$$

Any copper(II) salt in solution will react with sodium hydroxide solution in this way. For example, with copper(II) sulfate solution, the full equation is:

$$CuSO_4(aq) + 2NaOH(aq) \rightarrow Cu(OH)_2(s) + Na_2SO_4(aq)$$



▲ Figure 18.9 The blue precipitate of copper(II) hydroxide.

### THE TEST FOR IRON(III) IONS

**!**

Safety Note: Wear eye protection for all these tests and avoid skin contact with the reactants and products.

Formation of an orange-brown precipitate shows the presence of iron(III) ions ($Fe^{3+}$). The precipitate is iron(III) hydroxide.

$$Fe^{3+}(aq) + 3OH^-(aq) \rightarrow Fe(OH)_3(s)$$

Any iron(III) compound in solution will give this precipitate. An example of a full equation might be:

$$FeCl_3(aq) + 3NaOH(aq) \rightarrow Fe(OH)_3(s) + 3NaCl(aq)$$

Notice how much more complicated the full equations for these reactions are. They also hide what is going on. Use ionic equations for precipitation reactions wherever possible.



▲ Figure 18.10 The orange-brown precipitate of iron(III) hydroxide.

## THE TEST FOR IRON(II) IONS

**KEY POINT**

The green precipitate darkens on standing and turns orange-brown around the top of the tube. This is due to the iron(II) hydroxide being oxidised to iron(III) hydroxide by the air.

**KEY POINT**

You must carry out these three tests on a solution of the salt you are testing. If you tested the solid you would not be able to see whether a precipitate had been formed because there is already solid present.

Formation of a green precipitate shows the presence of iron(II) ions ($Fe^{2+}$). The precipitate is iron(II) hydroxide.

$$Fe^{2+}(aq) + 2OH^-(aq) \rightarrow Fe(OH)_2(s)$$

This could be the result of reacting, say, iron(II) sulfate solution with sodium hydroxide solution.

$$FeSO_4(aq) + 2NaOH(aq) \rightarrow Fe(OH)_2(s) + Na_2SO_4(aq)$$

**!**

Safety Note: Wear eye protection for all these tests and avoid skin contact with the reactants and products.



▲ Figure 18.11 The green precipitate of iron(II) hydroxide.

## THE TEST FOR AMMONIUM IONS

**KEY POINT**

Recognising gases by smelling them has to be done with great care. In this case, there is usually so little ammonia gas present in the cold that it is safe to smell if you take the normal precautions. You shouldn't, however, make any attempt to smell the mixture when it is warm.
Safety Note: Do not attempt to smell the ammonia gas if you have a breathing problem such as asthma.

This test shows the presence of an ammonium salt. Sodium hydroxide solution reacts with ammonium salts (either solid or in solution) to produce ammonia gas. In the cold, there is just enough ammonia gas produced for you to be able to smell it. If you warm it, you can test the gas coming off with a piece of damp red litmus paper or universal indicator paper. Ammonia is alkaline and turns the litmus paper/universal indicator blue. The ionic equation is:

$$NH_4^+(aq) + OH^-(aq) \rightarrow NH_3(g) + H_2O(l)$$

This test works on a sample of a solid or a solution as you are not looking for a precipitate.

A full equation for adding sodium hydroxide solution to solid ammonium chloride is:

$$NH_4Cl(s) + NaOH(aq) \rightarrow NaCl(aq) + NH_3(g) + H_2O(l)$$

Table 18.1 A summary of the reactions of positive ions with sodium hydroxide solution

| Ion | Result of adding NaOH(aq) | Name of precipitate formed | Formula of precipitate |
|---|---|---|---|
| $Cu^{2+}$ | blue precipitate | copper(II) hydroxide | $Cu(OH)_2$ |
| $Fe^{2+}$ | green precipitate | iron(II) hydroxide | $Fe(OH)_2$ |
| $Fe^{3+}$ | orange-brown precipitate | iron(III) hydroxide | $Fe(OH)_3$ |
| $NH_4^+$ | ammonia gas produced | no precipitate | |

## TESTING FOR CARBONATES ($CO_3^{2-}$)

**EXTENSION WORK**

The ionic equation shows any carbonate reacting with any acid.

$CO_3^{2-}(s) + 2H^+(aq) \rightarrow CO_2(g) + H_2O(l)$

Add a little dilute hydrochloric acid to your sample of salt and look for fizzing/bubbles of gas. Fizzing indicates that a gas is given off. You can test the gas by bubbling it through limewater to show that it is carbon dioxide.

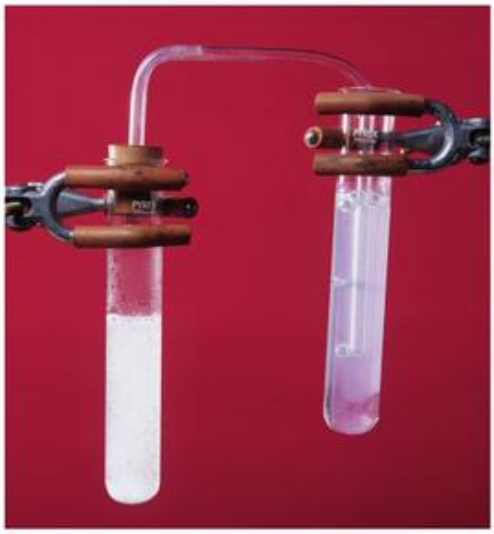
Most carbonates are insoluble in water and so you will usually be doing this test on a sample of a solid. Soluble carbonates are sodium carbonate, potassium carbonate and ammonium carbonate, and the test also works on solutions of these.

**HINT**

You can use any acid to do this test, but you have to be careful, especially with sulfuric acid, as some acid–carbonate combinations can produce an insoluble salt that coats the solid carbonate and stops the reaction. The test works well with dilute hydrochloric acid or dilute nitric acid.

For example, using zinc carbonate and dilute nitric acid:

$$ZnCO_3(s) + 2HNO_3(aq) \rightarrow Zn(NO_3)_2(aq) + CO_2(g) + H_2O(l)$$



▲ Figure 18.12 We can test to see if carbon dioxide is produced by bubbling the gas through limewater (in the tube on the right) and looking to see whether the limewater goes cloudy/milky (white precipitate of calcium carbonate formed).

**!**

Safety Note: Wear eye protection for all these tests and avoid skin contact with the reactants and products.

## TESTING FOR SULFATES ($SO_4^{2-}$)

**KEY POINT**

You could also use nitric acid and barium nitrate solution in the same way.

Make a solution of your suspected sulfate in distilled (pure) water, add some *dilute hydrochloric acid* and then add some *barium chloride solution*. A sulfate will produce a white precipitate of barium sulfate.

$$Ba^{2+}(aq) + SO_4^{2-}(aq) \rightarrow BaSO_4(s)$$

You add acid to the solution to react with and remove other anions which might also produce white precipitates when you add the barium chloride solution. For example, if you didn't add acid, you would also obtain a white precipitate if there were carbonate ions present. This is because barium carbonate is also white and insoluble. The acid reacts with the carbonate ions and removes them. It is important to remember that the acid must be added *before* the barium chloride solution.

You must never acidify the solution with sulfuric acid because sulfuric acid contains sulfate ions. If you add those, you are bound to get a precipitate of barium sulfate because you have added sulfate ions.

## TESTING FOR CHLORIDES ($Cl^-$), BROMIDES ($Br^-$) AND IODIDES ($I^-$)

Make a solution of your suspected chloride, bromide or iodide and add enough *dilute nitric acid* to make it acidic. Then add some *silver nitrate solution*.

**!**

Safety Note: Wear eye protection for all these tests and avoid skin contact with the reactants and products.

A *white* precipitate (of silver chloride) shows the presence of chloride ions:

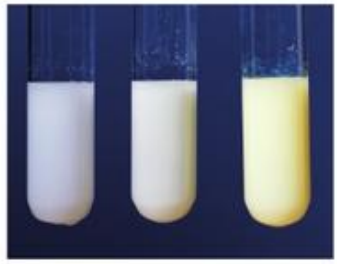
$$Ag^+(aq) + Cl^-(aq) \rightarrow AgCl(s)$$

A *cream* precipitate (of silver bromide) shows the presence of bromide ions:

$$Ag^+(aq) + Br^-(aq) \rightarrow AgBr(s)$$

A *yellow* precipitate (of silver iodide) shows the presence of iodide ions:

$$Ag^+(aq) + I^-(aq) \rightarrow AgI(s)$$



▲ Figure 18.13 Precipitates of silver chloride, silver bromide and silver iodide

**KEY POINT**

All of these precipitates tend to discolour to greys and pale purples on exposure to light.

**KEY POINT**

The 1– ions formed by the halogens (Group 7) are collectively called *halide ions*. Chloride, bromide and iodide ions are all halide ions.

The acid is added to react with and remove other anions which might also produce precipitates with silver nitrate solution, for example carbonate and hydroxide ions. You must not use hydrochloric acid (or hydrobromic acid or hydroiodic acid) as you will be adding halide ions and you are certain to obtain a precipitate.

Table 18.2 A summary of the tests for negative ions (anions)

| Ion | Test | Positive result | Product of reaction |
|---|---|---|---|
| $CO_3^{2-}$ | add dilute hydrochloric acid then bubble any gas produced through limewater | fizzing, limewater goes cloudy (white precipitate) | $CO_2$ $CaCO_3$ (with limewater) |
| $SO_4^{2-}$ | add dilute hydrochloric acid followed by barium chloride solution | white precipitate | $BaSO_4$ |
| $Cl^-$ | add dilute nitric acid followed by silver nitrate solution | white precipitate | AgCl |
| $Br^-$ | add dilute nitric acid followed by silver nitrate solution | cream precipitate | AgBr |
| $I^-$ | add dilute nitric acid followed by silver nitrate solution | yellow precipitate | AgI |

## CHAPTER QUESTIONS

SKILLS CRITICAL THINKING

1 Name the gas being described in each of the following cases.

a A green gas that bleaches damp litmus paper.

b A gas that dissolves readily in water to produce a solution with a pH of about 11.

c A gas that produces a white precipitate with limewater.

d A gas that pops when a lighted splint is placed in it.

e A gas that relights a glowing splint.

SKILLS EXECUTIVE FUNCTION

2 Describe in detail how you would carry out the following tests. In each case state whether you would carry out the test on the solid or a solution, and describe what you would expect to see happen.

a A flame test for lithium ions in lithium chloride.

b A test for ammonium ions in ammonium sulfate.

c A test for sulfate ions in ammonium sulfate.

d A test for carbonate ions in calcium carbonate.

e A test for iodide ions in potassium iodide.

3 **A** is an orange solid, which dissolves in water to give an orange solution. When sodium hydroxide solution is added to a solution of **A**, an orange-brown precipitate, **B**, is formed. Adding dilute nitric acid and silver nitrate solution to a solution of **A** gives a white precipitate, **C**.

SKILLS CRITICAL THINKING

a State the names of **A**, **B** and **C**.

SKILLS PROBLEM SOLVING

b Write equations (full or ionic) for the reactions producing **B** and **C**.

SKILLS PROBLEM SOLVING, CRITICAL THINKING (6)

4 **D** is a green crystalline solid that dissolves in water to give a very pale green solution. Addition of sodium hydroxide solution to a solution of **D** produces a green precipitate, **E**, which turns orange-brown around the top after standing in air. Addition of dilute hydrochloric acid and barium chloride solution to a solution of **D** gives a white precipitate, **F**.

a State the names of **D**, **E** and **F**.

SKILLS PROBLEM SOLVING (9)

b Write equations (full or ionic) for the reactions producing **E** and **F**.

(6)

5 **G** is a colourless crystalline solid which reacts with dilute nitric acid to give a colourless solution, **H**, and a colourless, odourless gas, **I**, which turns limewater milky. **G** has a lilac flame colour.

SKILLS CRITICAL THINKING

a State the names of **G**, **H** and **I**.

SKILLS PROBLEM SOLVING (9)

b Write an equation (full or ionic) for the reaction between **G** and dilute nitric acid.

(6)

6 **J** is a colourless solid. When it is warmed with sodium hydroxide solution it produces a gas, **K**, which turns damp universal indicator paper blue. When nitric acid, followed by silver nitrate solution, is added to a solution of **J**, a cream precipitate of **L** is formed.

SKILLS CRITICAL THINKING

a State the names of **J**, **K** and **L**.

SKILLS PROBLEM SOLVING (9)

b Write an ionic equation for the formation of the cream precipitate.

SKILLS EXECUTVE FUNCTION (5)

7 A student has found that her sample of potassium nitrate is contaminated with small amounts of a green solid. She picks out a small piece of the green solid and finds that it is insoluble in water.

SKILLS CRITICAL THINKING (6)

a Describe how you would make a pure sample of potassium nitrate from the impure mixture.

b The student believes that the green solid could be copper(II) carbonate. Describe a series of tests that the student could use to confirm this.

(5)

8 A student finds a bottle containing a colourless liquid in the laboratory.

a Describe a test to show that the bottle contains water.

(4)

b Describe a test to show whether it is pure water.

SKILLS EXECUTIVE FUNCTION, REASONING (7)

c The student carried out the test in **b** and it showed that the water was not pure. He then carried out some more tests to try to identify the solution. The following results were obtained:

| Test | Result |
|---|---|
| add sodium hydroxide solution | no visible reaction but the test-tube became warm |
| add dilute hydrochloric acid | no reaction |
| add dilute hydrochloric acid followed by barium chloride solution | no reaction |
| add nitric acid followed by silver nitrate solution | white precipitate |

Explain what could be present in the bottle and how the student could confirm this.